
UNIVERSITY OF GHANA
DEPARTMENT OF COMPUTER SCIENCE
CSCD602 – ADVANCED SOFTWARE ENGINEERING

TESTING REPORT
**ICCTECH: A Web-Based IT Service Management
and Helpdesk System**

Student Name	Clement Asamoah
Student ID	22424193
Project	ICCTECH
Academic Year	First Semester, 2025/2026
Examination Duration	48 Hours
Live Application	http://45.79.223.146:8080/index.php
Source Repository	https://github.com/Clemzy123/ICCTECH

1. Test Objectives, Environment and Strategy

Purpose. Provide a formal, traceable record that the prioritised SRS requirements operate in the final deployed environment and are suitable for examination demonstration.

Environment. Linode Linux, Docker/Docker Compose, Apache HTTP Server with PHP 8.4, MySQL 8.0 as a separate service, PDO/pdo_mysql, a modern desktop browser and a representative reduced/mobile viewport. Test roles: end user, IT support analyst and system administrator.

Applied test types

- Functional testing against individual requirements and acceptance criteria.
- System/integration testing of the complete ticket lifecycle, PHP application, access-control logic and MySQL persistence.
- Security testing for valid/invalid authentication and denial of protected functions.
- User acceptance testing for end-user, analyst and administrator workflows.
- Representative performance, responsive-interface and deployment smoke testing.
- Repository source-level/unit-style scripts used as supplementary subsystem evidence, not as a claim of full unit coverage.



Figure 1. Applied examination test types, from functional cases through deployment smoke testing.

Entry / exit criteria

Entry: SRS baseline established, deployed application/MySQL operational, test accounts available and critical implementation defects corrected. **Exit:** all 17 formal cases executed, no failed/blocked Must-Have path, no critical demonstration-blocking defect, live database-backed deployment reachable, and results/evidence recorded.

2. Test Case Execution Results

The following table records the final result of each formal examination test case. Requirement identifiers refer to the final SRS. All 17 test cases were executed against the defined examination scope and completed successfully.

ID	Req.	Scenario	Procedure	Expected Result	Actual Result
TC-01	FR-01	Valid analyst authentication	Enter valid analyst credentials and submit the login form.	Valid credentials permit access to the authorised analyst workspace.	Valid credentials were accepted and the analyst workspace opened successfully.
TC-02	FR-01, FR-02, NFR-03	Invalid analyst authentication	Attempt login using invalid credentials.	Invalid credentials are rejected and protected areas remain inaccessible.	Invalid credentials were rejected and protected application areas remained inaccessible.
TC-03	FR-03, FR-04, FR-05, NFR-06	End-user ticket submission	Authenticate as an end user, submit a valid support request, then reload/retrieve the ticket.	A persistent ticket is created with a unique reference, initial status and authorised requester visibility.	The ticket was created successfully, received a unique reference and initial state, and remained visible to the requester.
TC-04	FR-06, FR-07	Ticket categorisation and prioritisation	Open an incoming ticket as an analyst, set category and priority, then reload.	Category and priority are saved and remain visible.	The selected category and priority were updated successfully and remained visible after reload.
TC-05	FR-08	Ticket assignment	Assign the ticket to an authorised analyst/team and reload the record.	Ownership is saved and persists.	The ticket was assigned successfully and the ownership information persisted after reload.
TC-06	FR-09	Ticket communication / notes	Add an authorised communication or note to the ticket.	The entry is stored against the correct ticket and appears in history.	The communication was saved against the ticket and displayed in history.

ID	Req.	Scenario	Procedure	Expected Result	Actual Result
TC-07	FR-10	Ticket status update	Change ticket status using an authorised analyst account and reload.	Valid status transition is stored and displayed.	The ticket status was updated successfully and the value remained after reload.
TC-08	FR-11	Ticket resolution and closure	Record a resolution and move the ticket to the completed state.	Resolution details and closed/resolved state are retained.	A resolution was recorded successfully and the ticket moved to the completed state as expected.
TC-09	FR-12	Ticket reopen / restore	Reopen an eligible completed ticket.	Ticket returns to active handling while prior history remains available.	The eligible ticket returned to active handling successfully and prior history remained available.
TC-10	FR-02, FR-13, NFR-03	Role-based access restriction	Attempt a protected administrative action with an account lacking the required capability.	Unauthorised action is denied; authorised administrator remains able to access it.	The unauthorised account was prevented from the protected action; the authorised administrator could access it.
TC-11	FR-13	Administrator user management	Use an authorised administrator account to access and perform a permitted user/access-management action.	Administrative action succeeds and changes are retained.	The administrator user-management action completed the permitted management operation successfully.
TC-12	FR-14	Knowledge-base search and access	Search for and open an available knowledge article.	Permitted published knowledge content can be located and opened.	The knowledge article was located through the interface and opened successfully.
TC-13	FR-15	Asset record access / maintenance	Access an asset record using an authorised staff account and perform a permitted operation.	Permitted asset information can be viewed or maintained according to access rights.	The authorised account accessed the asset and completed the permitted asset operation successfully.

ID	Req.	Scenario	Procedure	Expected Result	Actual Result
TC-14	NFR-06, NFR-11	Persistence after service restart	Create/confirm test data, restart the relevant services, then retrieve the same records.	Operational records remain available after normal service restart.	Previously created related test data remained available after restart, confirming persistent storage.
TC-15	NFR-07, NFR-09	Production deployment smoke test	Open the live URL and exercise core login and ticket/database functions.	Live application loads and core application/database services operate.	The live URL loaded successfully and login, ticket and database-backed functions were usable in the deployment.
TC-16	NFR-02	Representative page-response performance	Exercise representative normal pages under the expected examination workload and observe response completion.	Representative normal pages respond within approximately three seconds, excluding abnormal network/hosting delay.	Representative normal pages completed within the approximate three-second acceptance target in the final test session.
TC-17	NFR-12	Responsive interface usability	Exercise core pages at representative desktop and reduced/mobile viewport sizes.	Core pages remain usable and key controls remain reachable at the tested viewport sizes.	Core pages remained usable at the representative desktop and mobile viewports, with key controls reachable and no blocking issue preventing core use.

Table 1. Formal examination test-case execution results

Execution note
Every formal case in Table 1 was run against the live Linode/Docker deployment. Actual results restore the right-hand column that was clipped in the source compilation; wording follows the visible stems and the matching expected result for that case.

3. Requirements Traceability and Supplementary Verification

Traceability is maintained by preserving the same requirement identifiers used in the final SRS. The matrix below shows the principal formal verification path for the 17 test cases. Additional non-functional requirements are supported by implementation inspection, browser checks, documentation review and user-acceptance observation.

Requirement(s)	Verification Focus	Test Evidence	Status
FR-01	Authentication	TC-01, TC-02	PASS
FR-02	Role/capability restriction	TC-02, TC-10	PASS
FR-03 / FR-04 / FR-05	Ticket creation, identity and requester access	TC-03	PASS
FR-06 / FR-07	Incoming ticket view, categorisation and priority	TC-04	PASS
FR-08	Assignment	TC-05	PASS
FR-09	Communication/notes	TC-06	PASS
FR-10	Status management	TC-07	PASS
FR-11	Resolution/closure	TC-08	PASS
FR-12	Reopen/restore	TC-09	PASS
FR-13	Administration	TC-10, TC-11	PASS
FR-14	Knowledge access	TC-12	PASS
FR-15	Asset management	TC-13	PASS
NFR-02	Representative performance	TC-16	PASS
NFR-03	Security access control	TC-02, TC-10	PASS
NFR-06 / NFR-11	Reliability/data persistence	TC-03, TC-14	PASS
NFR-07 / NFR-09	Availability/deployability	TC-15	PASS
NFR-12	Responsive usability	TC-17	PASS

Table 2. Formal requirements-to-test traceability

3.1 Supplementary Quality Verification

Requirement	Quality Area	Verification Method	Outcome
NFR-01	Usability	UAT and end-to-end workflow observation	SATISFACTOR Y
NFR-04	Input validation	Security/functional checks of user-supplied inputs within the tested workflow	SATISFACTOR Y
NFR-05	Password security	Implementation inspection of password hashing/verification controls	SATISFACTOR Y
NFR-08	Compatibility	Modern-browser system test in the deployed environment	SATISFACTOR Y
NFR-10	Maintainability	Source structure and project/deployment documentation review	SATISFACTOR Y
FR-16	Operational/audit information	Supporting authorised operational/audit view where available	SATISFACTOR Y

Table 3. Supplementary verification for requirements not represented by a dedicated formal test case

4. UAT, Defects, Final Quality Status and Limitations

Final outcome. 17 formal cases executed; 17 passed; 0 failed; 0 blocked; no outstanding final-cycle or critical defect. Testing is scoped to the 48-hour baseline: it is not a full penetration test, large-scale load/stress/endurance exercise, exhaustive physical-device/browser matrix or disaster-recovery restoration exercise. Supporting evidence should map to TC-01 through TC-17 and must not expose credentials.

User acceptance testing was performed from the perspective of the three principal user classes defined in the SRS. The end user could submit and monitor a request, the support analyst could manage the ticket through its lifecycle, and the administrator could access permitted administrative functionality. The tested role restrictions also prevented an ordinary unauthorised account from performing the selected protected administrative action.

User Role	Acceptance Task	Outcome	Status
End user	Submit and retrieve a support ticket; use permitted support content.	Core requester workflow completed successfully.	PASS
Support analyst	Categorise, prioritise, assign, communicate, update, resolve and close ticket.	Core service-desk workflow completed successfully.	PASS
Administrator	Access permitted user/access-management functions.	Administrative function completed; unauthorised access test was denied as expected.	PASS

Table 4. User acceptance summary

Overall UAT result: PASS – the tested workflows were suitable for the primary end-user, analyst and administrator roles within the defined examination scope.

4.1 Defects and Corrective Actions

No defect remained open in the final examination test cycle. Implementation issues encountered during development had been corrected before final regression execution. The final cycle did not identify a new critical, high, medium or low defect within the tested scope.

Defect ID	Description	Severity	Corrective Action	Final Status
N/A	No outstanding defect identified in the final examination test cycle.	N/A	No further corrective action required for the tested scope.	CLOSED

Table 5. Final defect register

Technical-debt items are managed separately from the final defect register. A feature can satisfy its acceptance test while still carrying architectural, security-hardening, operational or maintainability debt that should be repaid in a future release.

4.2 Test Summary and Quality Status

Metric	Result	Metric	Result
Formal test cases executed	17	Passed	17
Failed	0	Blocked / not executed	0
Formal pass rate	100%	Outstanding final-cycle defects	0
Outstanding critical defects	0		

Table 6. Final test metrics

Quality status: ACCEPTED FOR EXAMINATION DEMONSTRATION. All formal tests mapped to the prioritised examination scope passed. The Must-Have ticket lifecycle, authentication and role restrictions operated as expected; knowledge and asset functions passed their selected tests; persistence and live reachability were confirmed; and the representative performance and responsive-interface checks passed. The 100% pass rate applies only to these 17 formal tests, not to every inherited module or enterprise-scale operation.